



NATIONAL JUNIOR COLLEGE
SH 1 PROMOTIONAL EXAMINATIONS

Higher 2

CANDIDATE
NAME

SUBJECT
CLASS

REGISTRATION
NUMBER

CHEMISTRY

9729/02

Paper 2 Structured Questions and Free Response Questions

Tues 1 Oct 2019

Additional Materials: Data Booklet
Writing Paper

2 hours 30 min

READ THE INSTRUCTIONS FIRST		For Examiner's Use	
Write your name, subject class and registration number on all the work you hand in. Write in dark blue or black ink on both sides of the paper. You may use a soft pencil for any diagrams or graphs. Do not use paper clips, highlighters, glue or correction fluid. Section A: Structured Questions Answer ALL questions on the question paper. Section B: Answer ALL questions from this section in the Answer Booklet. The use of an approved calculator is expected where appropriate. A Data Booklet is provided. The number of marks is given in brackets [] at the end of each question or part question. Appropriate significant figures and units are expected for final numerical answers. Submit answers to Section A and Section B separately.		Section A:	
		1	/4
		2	/12
		3	/17
		4	/11
		5	/15
		Section B:	
		1	/22
		2	/19
		Penalty	
		s.f.	
		units	
		Total	/100

This document consists of **18** printed pages.

Section A: Structured Questions [Total: 59 marks]Answer **ALL** questions in the space provided.

- 1 (a) A sample of lead contains four stable isotopes with the following percentage abundances.

Isotope	Percentage abundance / %
^{204}Pb	1.4
^{206}Pb	24.1
^{207}Pb	22.1
^{208}Pb	a

Determine the value of **a**.

Hence calculate the relative atomic mass of lead in the sample. Give your answer to two decimal places. [2]

$$\% \text{ abundance of } ^{208}\text{Pb}, a = 100 - 1.4 - 24.1 - 22.1 = 52.4\%$$

Relative atomic mass of lead

$$= (204 \times 1.4\%) + (206 \times 24.1\%) + (207 \times 22.1\%) + (208 \times 52.4\%)$$

$$= 207.24 \text{ (2 d.p.)}$$

- (b) Beams of charged particles are deflected by an electric field. The particles are travelling at the same speed through an electric field of constant strength. In a particular experimental set-up, protons are deflected through an angle of $+26^\circ$.

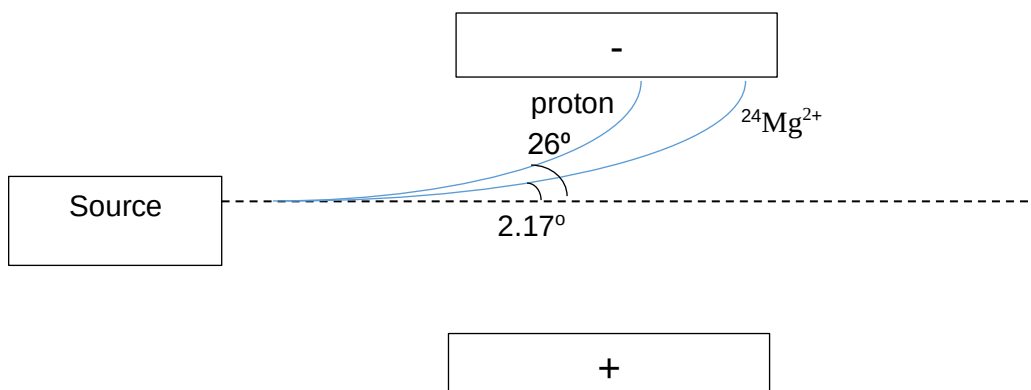
A beam of $^{24}\text{Mg}^{2+}$ ions was subjected to an identical set of experimental conditions. Calculate the angle deflected, giving your answer to 2 decimal places.

$$\text{Angle of deflection for } \text{H}^+ = 1/1 \times 26^\circ = 26^\circ$$

$$\text{Angle of deflection for } ^{24}\text{Mg}^{2+} = 2/24 \times 26^\circ = 2.17^\circ$$

Draw the paths taken by both the protons and $^{24}\text{Mg}^{2+}$ ions in the diagram below. [2]

Correct relative angles, labelled, pointing to negative plate



- 2 (a) (i) Butane is used as a propellant in hair spray to create an aerosol mist of liquid particles. The bottle of hair spray has an initial temperature of 27 °C. When the hair spray is used up, only butane gas remained. The internal pressure in the bottle decreased to 138 kPa.

Calculate the new internal pressure when the bottle reaches 50 °C on a hot afternoon.
[1]

Since V is constant

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{138}{273 + 27} = \frac{P_2}{273 + 50}$$

$$P_2 = 149 \text{ kPa}$$

- (ii) As a safety precaution, the propellant has to be completely expelled from the bottle prior to incineration. Explain qualitatively why this precaution is necessary. [1]

When it is incinerated, it will be subjected to **high temperatures**. As the volume of the bottle is constant, **pressure** of butane gas **increases**, **explosion** will occur.

- (b) One means of measuring toxicity is using LD, which stands for "Lethal Dose". LD₅₀ is the amount of a material which causes the death of 50% of a group of test animals. LD₅₀ value is expressed as the mass of a chemical administered per kg body mass of a test animal.

Another means of measuring toxicity is using LC, which stands for "Lethal Concentration". The concentration of the chemical in air that kills 50% of the test animals during the observation period is the LC₅₀ value.

The table below shows the values for the LD₅₀ and LC₅₀ along with the toxicity ratings.
(1 g = 1000 mg)

Toxicity rating	Commonly used term	LD ₅₀ : Oral (mg kg ⁻¹)	LC ₅₀ : Inhalation (ppm)
1	Extremely Toxic	1 or less	10 or less
2	Highly Toxic	>1 – 50	11 – 100
3	Moderately Toxic	51 – 500	101 – 1000
4	Slightly Toxic	501 – 5000	1001 – 10,000
5	Practically Non-toxic	5001 – 15,000	10,001 – 100,000

- (i) 4.45×10^{-4} mol of a toxic compound, C_4H_5NO , was found to cause death in 50 % of test animals weighing 1 kg.
Calculate LD_{50} of the compound and state its toxicity rating. [2]

$$\begin{aligned}\text{Mass of compound} &= 4.45 \times 10^{-4} \times (48 + 5 + 14 + 16) \\ &= 0.03694 \text{ g}\end{aligned}$$

Hence, $LD_{50} = 36.9 \text{ mg/kg}$ with correct units

The compound is **highly toxic** / The toxicity rating is **2**.

- (ii) Phosphine gas, PH_3 , is widely used in the semi-conductor industry as a dopant. The concentration of a small quantity of gas is usually expressed in parts per million (ppm) as shown below:

$$\text{Concentration in ppm} = \frac{\text{volume of gas}}{\text{volume of air}} \times 10^6$$

Given that LC_{50} for $PH_3(g)$ is 200 mg m^{-3} at room temperature and pressure, convert LC_{50} to ppm and determine its toxicity rating. [2]

$$\begin{aligned}\text{Volume of } PH_3 &= \frac{200 \times 10^{-3}}{34.0} \times 24 \text{ dm}^3 \\ &= 0.141 \text{ dm}^3\end{aligned}$$

$$\begin{aligned}LC_{50} &= \frac{0.141 \times 10^{-3}}{1} \times 10^6 \\ &= 141 \text{ ppm}\end{aligned}$$

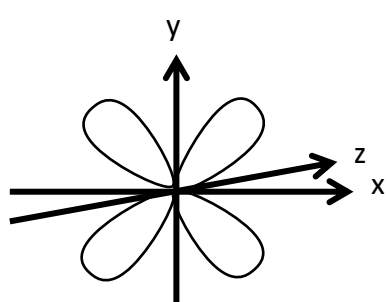
PH_3 is **moderately toxic** / The toxicity rating is **3**.

- (c) (i) The name of the element, chromium, means colour in Greek because many chromium compounds are intensely coloured.

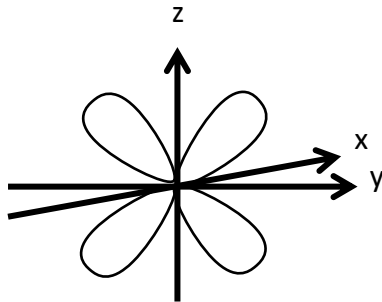
State the full electronic configuration of a chromium atom in its ground state. [1]



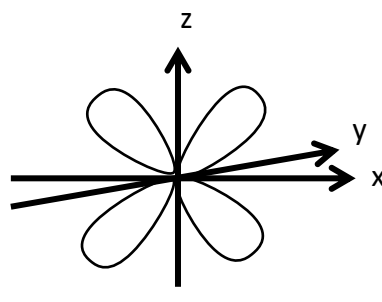
- (ii) Draw and label the shapes of all the d orbitals, showing the axes clearly. [2]



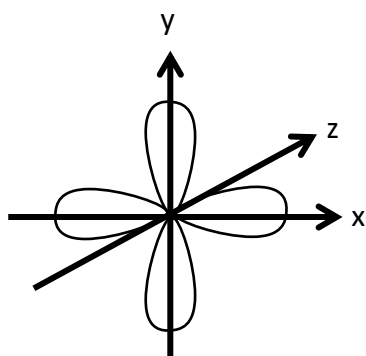
d_{xy} orbital



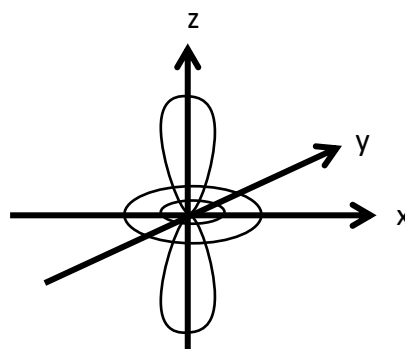
d_{yz} orbital



d_{xz} orbital



$d_{x^2-y^2}$ orbital



d_{z^2} orbital

- (d) The successive ionisation energy of any element can be determined in the laboratory.

- (i) Write an equation for the fifth ionisation energy for sulfur. [1]



- (ii) Hence, explain the difference in the fifth ionisation energies between sulfur and chlorine. [2]

The electron to be removed from Cl^{4+} is from **3p subshell while that for S^{4+} is from 3s subshell.**

3p subshell is further away from nucleus/ at higher energy than 3s subshell.

This outweighs the higher nuclear charge of Cl.

Therefore, **nuclear attraction for the 3p electron in Cl^{4+} is weaker, less energy is required to remove it.**

Hence, **5th IE of Cl is lower** than S.

- 3 Hematite is a common iron oxide with the formula Fe_2O_3 . It is a very important naturally occurring compound that finds widespread use as a heterogeneous catalyst.

(a) (i) Describe how a heterogeneous catalyst speeds up gaseous reactions. [3]

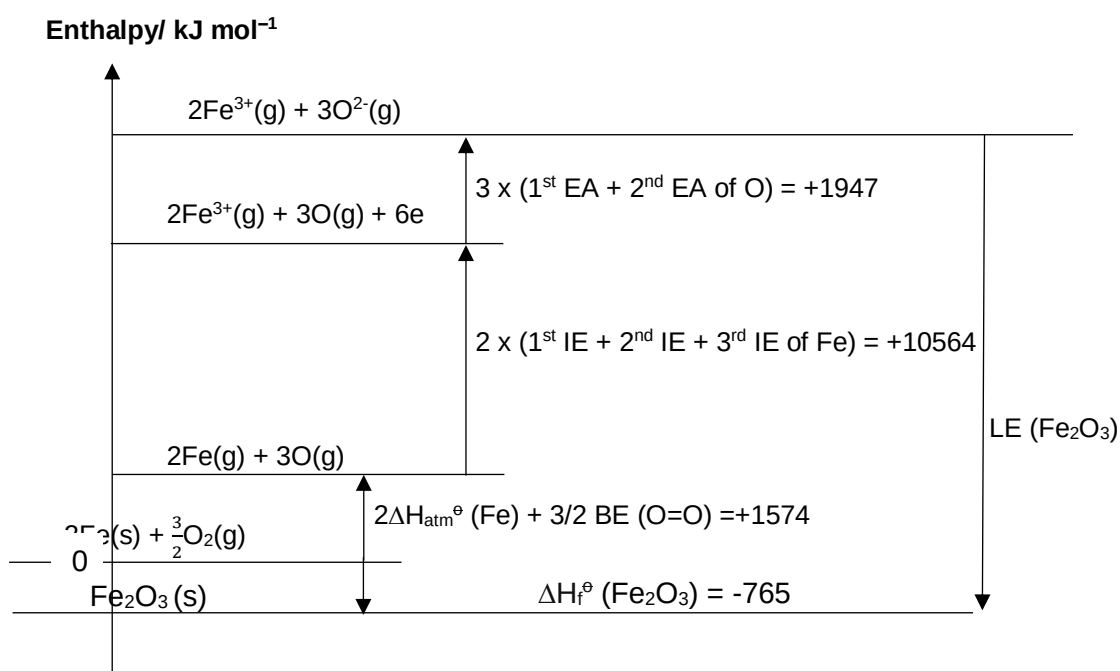
Adsorption of reactant particles onto the surface of the catalyst. **Weak interactions form between the reactant and catalyst.**

Reaction at the surface occurs at a faster rate as **reactant molecules are brought closer together** and **existing interactions in the reactant molecules are weakened**, thus lowering activation energy

Desorption of products from the catalyst surface. Catalyst is regenerated.

- (ii) Using the relevant values from the *Data Booklet* and information provided in the table below, draw a Born-Haber cycle for the formation of Fe_2O_3 and use it to calculate the lattice energy of Fe_2O_3 . [4]

Enthalpy change of atomisation of iron	/ kJ mol^{-1}	+ 415
Enthalpy change of formation of Fe_2O_3	/ kJ mol^{-1}	– 765
First electron affinity of oxygen	/ kJ mol^{-1}	– 141
Second electron affinity of oxygen	/ kJ mol^{-1}	+ 790



$$\begin{aligned}\text{LE}(\text{Fe}_2\text{O}_3) &= (-1947) + (-10564) + (-1574) + (-765) \\ &= 14850 \text{ kJ mol}^{-1} \text{ or } 14900 \text{ kJ mol}^{-1}\end{aligned}$$

- (iii) From knowledge of the distances between the cations and anions in the crystal structures and the charge on each ion, it is also possible to calculate theoretical values for lattice energies.

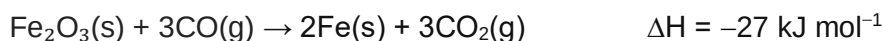
Given that the theoretical value of lattice energy for Fe_2O_3 is $-14774 \text{ kJ mol}^{-1}$, account for the discrepancy between your answer in (ii) and the theoretical value. [2]

The theoretical value is derived based on the assumption that cations and anions exist as discrete point charges with no electron density between them.

However, Fe^{3+} has high charge/size ratio, thus can distort the electron cloud of oxide, leading to covalent character in Fe_2O_3 .

Hence there is a discrepancy between the 2 values.

- iv) Fe_2O_3 can also react with CO to produce Fe.



Calculate $\Delta H_f^\ominus$ for $\text{CO}(\text{g})$, given that $\Delta H_f^\ominus$ for $\text{CO}_2(\text{g}) = -394 \text{ kJ mol}^{-1}$ and using suitable values from (a)(ii). [2]

$$\Delta H_r^\ominus = \sum n \Delta H_f^\ominus (\text{products}) - \sum m \Delta H_f^\ominus (\text{reactants})$$

$$-27 = (3 \times -394) - [(-765) + (3 \times \Delta H_f^\ominus \text{ for CO(g)})]$$

$$\Delta H_f^\ominus \text{ for CO(g)} = -130 \text{ kJ mol}^{-1}$$

(b) Fe^{2+} ions are also commonly used as a homogenous catalyst in reactions involving two anions.

(i) The reaction between peroxodisulfate ions, $\text{S}_2\text{O}_8^{2-}$ and iodide ions is very slow at room condition.

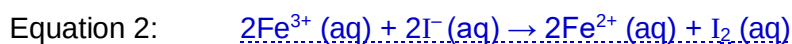
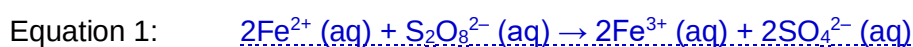
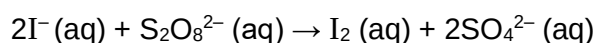
Explain why this is so.

[1]

$\text{S}_2\text{O}_8^{2-}$ and I^- are both negatively charged and hence energy is required to overcome the repulsion between them before the reaction can occur.

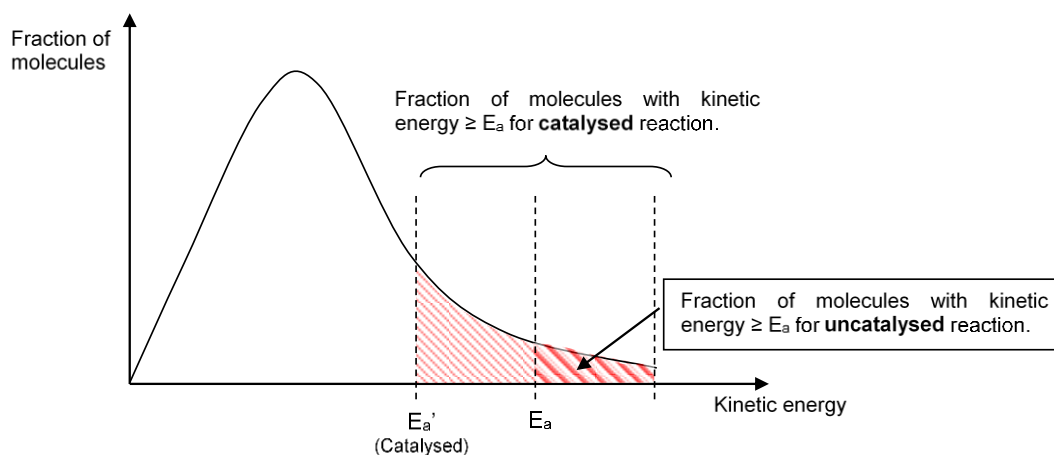
(ii) Write two equations to show how Fe^{2+} catalyses the reaction below.

[2]



(iii) With the aid of a Boltzman distribution curve, explain how Fe^{2+} ions catalyse the reaction between $\text{S}_2\text{O}_8^{2-}$ and iodide ions.

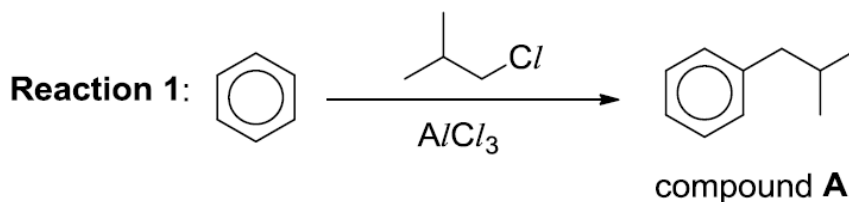
[3]



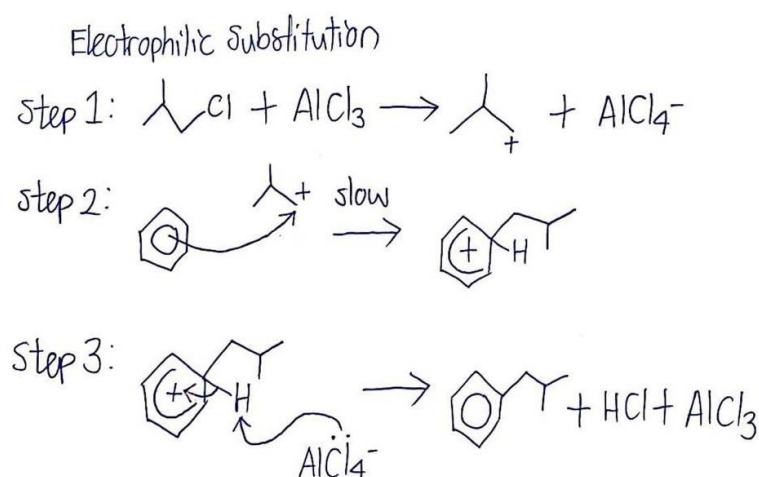
In the presence of catalyst, the reaction proceeds with an alternative reaction pathway of lower activation energy (E_a').

There are more particles with kinetic energy equal or greater than the lowered activation energy. This results in greater frequency of effective collisions. Hence rate of reaction increases.

- 4 | Compound **A** can be synthesised from benzene using aluminium chloride as shown in Reaction 1.



- (a) | Describe the mechanism for Reaction 1. You should show all relevant charges, lone pairs, dipoles and movement of electrons by curly arrows. [3]

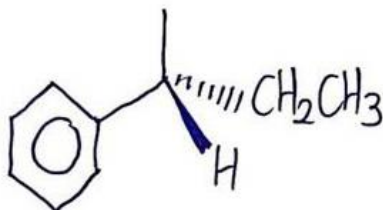


Name of mechanism

- Correct curly full arrows
- Slow step indicated
- Charges shown
- lone pairs shown
- Regeneration of AlCl_3
- HCl as product
- Correct electrophile
- Correct arenium carbocation

- (b) | Compound **B**, which contains a chiral centre and is an isomer of **A**, is also formed in Reaction 1. **B** is formed after the electrophile undergoes rearrangement through the movement of an alkyl group to an adjacent carbon atom bearing the positive charge.

- (i) | Give the structure of **B**. [1]

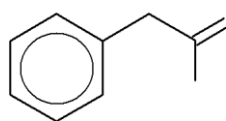


- (ii) Suggest the structure of the electrophile that gives rise to **B** and explain why the formation of this electrophile is favoured. [3]



The carbocation that forms **B**, as shown above, has 2 electron-donating alkyl groups while the carbocation that forms **A** has only 1 electron-donating alkyl group. The positive charge on carbocation that forms **B** is more dispersed and hence more stable as compared to that of carbocation that forms **A**, and formation of the electrophile that forms **B** is favoured.

- (c) Compound **C** has the structure as shown below.



Suggest a simple chemical test to distinguish between compounds **A** and **C**. [2]

Add Br_2 dissolved in CCl_4 / liquid Br_2 to both samples of **A** and **C**. If orange-red Br_2 decolourises, the compound is **C**. If orange-red Br_2 remains, the compound is **A**.

OR

Add aqueous Br_2 to both samples of **A** and **C**. If orange Br_2 decolourises, the compound is **C**. If orange Br_2 remains, the compound is **A**.

OR

Add cold KMnO_4 with NaOH(aq) to both samples of **A** and **C**. If purple KMnO_4 decolourises with formation of brown MnO_2 precipitate, the compound is **C**. If purple KMnO_4 remains, the compound is **A**.

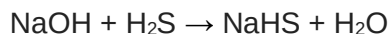
- (d) Suggest whether **C** would be soluble in water. [2]

C is insoluble in water. Energy liberated from forming temporary dipole-induced dipole interactions between **C** and water is insufficient to overcome temporary dipole-induced dipole interactions between **C**, and hydrogen bonding interactions between water molecules.

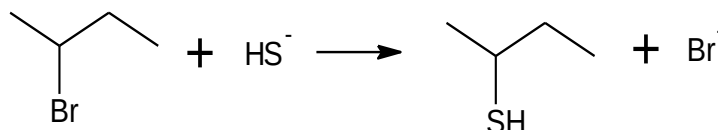
- 5 Many thiols have strong odors resembling that of garlic or rotten eggs. Thiols are used as odorants to assist in the detection of natural gas, and the "smell of natural gas" is due to the smell of the thiol used as the odorant.

The general formula of thiol is R-SH where **R** is an alkyl group.

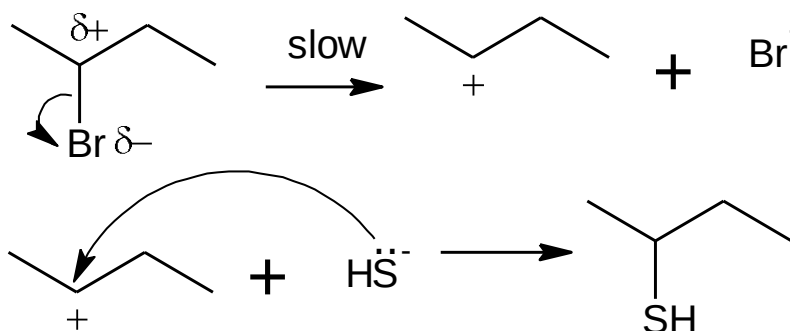
Thiols are formed when halogenoalkanes react with hydrogen sulfide ions via nucleophilic substitution. Hydrogen sulfide ions are generated as shown:



One such example is the reaction of 2-bromobutane with hydrogen sulfide ion via $\text{S}_{\text{N}}1$ mechanism.



- (a) Describe the mechanism of the above reaction, showing relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs. [2]



- (b) Explain why the resultant product is not optically active.

The carbocation is trigonal planar / planar. Therefore, HS^- has equal probability to attack C^+ / carbocation from top or bottom / either side / left or right of the plane. This results in equal amount of each enantiomer formed. The effect of rotation of plane-polarized light of one enantiomer cancels out that of the other.

- (c) What do you understand by the term *nucleophile*? Explain why hydrogen sulfide ion, HS^- , is a stronger nucleophile than hydrogen sulphide, H_2S . [2]

Nucleophile is a species which has an available lone pair to donate to another species. Due to the presence of **negative charge**, the lone pair on S is more available for donation.

- (d) The kinetics of the reaction between 1-bromobutane and hydrogen sulfide ions, HS^- , was studied and the following results were obtained.

Initial concentration of 1-bromobutane /mol dm ⁻³	Initial concentration of HS^- /mol dm ⁻³	Initial rate of reaction /mol dm ⁻³ s ⁻¹
0.1	0.1	1.5×10^{-5}
0.2	0.1	3.0×10^{-5}
0.3	0.2	9.0×10^{-5}
X	y	4.5×10^{-5}

- (i) Deduce the orders of reaction with respect to 1-bromobutane and HS⁻ ions. [2]

Compare experiment 1 and 2

When concentration of 1-bromobutane x 2, initial rate of reaction x 2, order of reaction with reaction wrt 1-bromobutane is 1.

Compare experiment 1 and 3.

$$\frac{\text{Rate 1}}{\text{Rate 3}} = \frac{k[0.1][0.1]^x}{k[0.3][0.2]^x}$$

$$\frac{1.5 \times 10^{-5}}{9.0 \times 10^{-5}} = \frac{k[0.1][0.1]^x}{k[0.3][0.2]^x}$$

$$x = 1$$

Order of reaction wrt HS⁻ is 1.

- (ii) Hence write the rate equation and determine the rate constant of the reaction. [2]

Rate = k[1-bromobutane][HS⁻] (1 mark)

From experiment 1,

$$1.5 \times 10^{-5} = k(0.1)(0.1)$$

$$k = 1.5 \times 10^{-3} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

- (iii) Suggest a set of values for x and y. [1]

x = 0.3, y = 0.1 or any other correct mathematical combination.

- (iv) Using the rate equation in (ii), suggest whether the reaction between 1-bromobutane and hydrogen sulfide ions, HS⁻, would be similar to the mechanism described in (a). [1]

From the rate equation, there is **1 mole of 1-bromobutane and 1 mol of HS⁻ in the slow step** and hence the mechanism is S_N2 and hence is different from mechanism described in (a).

Or

The **overall order of reaction as shown in the rate equation is 2** and hence the mechanism is S_N2 instead of S_N1.

(no ecf for explanation if the rate equation is deduced wrongly)

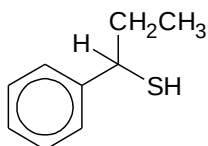
- (e) Two other thiols, **Y** and **Z**, have the following percentage composition by mass: C, 71%; H, 7.9%; S, 21.1%. Each molecule of **Y** and **Z** contains only 1 sulfur atom. Only **Z** is optically active and decolourises hot acidified potassium manganate(VII). However, **Y** does not decolourise hot acidified potassium manganate(VII).

[Note: S-H is inert to hot acidified potassium manganate(VII).]

- (i) Determine the empirical formula of **Z**, and suggest its structure. [2]

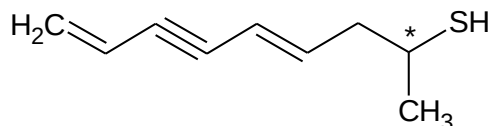
Element	C	H	S
Percentage	71	7.9	21.1
Mole	5.916	7.9	0.65937
simplest mole ratio	9	12	1

Empirical formula is $\text{C}_9\text{H}_{12}\text{S}$. Since there is only 1 sulfur atom, empirical formula is the same as molecular formula.

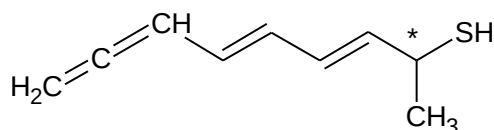


Structure of **Z** is

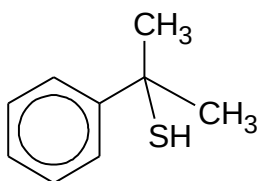
An acceptable answer is:



OR



- (ii) Suggest the structure of **Y**. [1]



Section B: Free Response Questions [Total: 41 marks]**Answer both questions in the Answer Booklet.**

- 1** The elements in Group 17, the halogens, and their compounds, show many similarities and trends in their properties. The table below shows some data for the elements, fluorine to iodine.

Element	Bond energy / kJ mol^{-1}	Standard enthalpy change of atomisation, $\Delta H^\circ_{\text{at}}$ / kJ mol^{-1}	Boiling point of element / K	Boiling point of hydrogen halide / K
Fluorine, F–F	158	79	85	293
Chlorine, Cl–Cl	244	122	238	188
Bromine, Br–Br	193	112	332	206
Iodine, I–I	151	107	457	238

- (a)** Explain the trend in the boiling points of the hydrogen halides, HF, HCl, HBr and HI. [3]

HF exhibit hydrogen bonding between molecules therefore requires more energy to overcome than temporary dipole induced dipole (tdid) that exist between molecules of the hydrogen halides down the group.

From HCl to HI, there is increase in electron cloud size that increases the ease of distortion leading to increasing strength of tdid between the respective molecules of the hydrogen halides. Hence, there is increase in the amount of energy needed to vaporise hydrogen halide from HCl to HI.

(b) 0.60 g of a gaseous halogen occupies 100 cm³ at 70 °C and 1 atm.

(i) Determine the relative molecular mass, M_r , of the halogen and deduce its identity. [2]

$$10^{-4} \times 101325 = \frac{0.60}{M} \times 8.31 \times (273 + 70)$$

$$M = 168.8 \text{ g mol}^{-1}$$

$$M_r = 168.8 \text{ (to 1 dp)}$$

Its likely to be bromine.

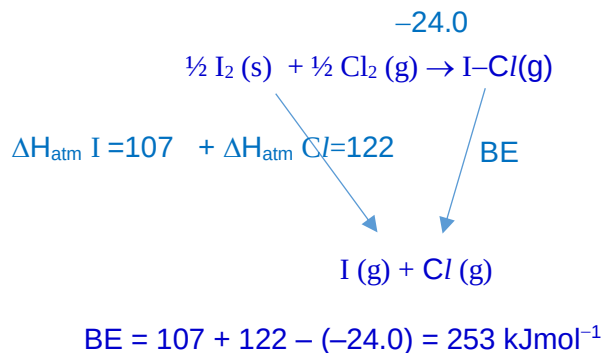
(ii) Explain why the M_r calculated in **(i)** is larger than expected. [1]

Volume measured is smaller than actual.

due to significant intermolecular attraction between the molecules.

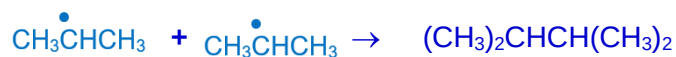
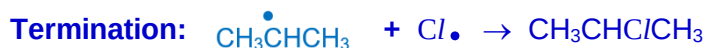
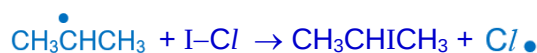
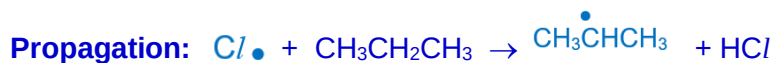
Since M_r is inversely proportional to volume, M_r will calculate to be bigger.

- (c) (i) The standard enthalpy change of formation of I-Cl(g) is $-24.0 \text{ kJ mol}^{-1}$.
Use this information and the table above to calculate the bond energy of I-Cl with a suitable energy cycle. [2]

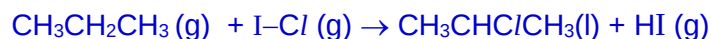


- (ii) Propane can react with I-Cl to form 2-iodopropane.
Describe the mechanism for formation of 2-iodopropane, showing curly arrows. [3]

Free radical substitution



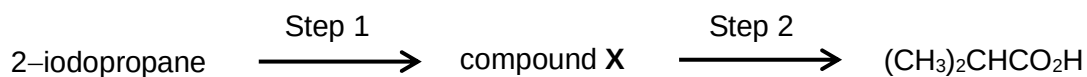
- (iii) Using relevant data from the *Data Booklet* and your answer in (i), verify that the formation of 2-chloropropane from the reaction described in (ii) is always non-spontaneous. [3]



$$410 + 253 - 340 - 299 = 24 \text{ kJmol}^{-1}$$

$\Delta n_{\text{gas}} = 1 - 2 < 0$, therefore there is a drop in entropy causing ΔS to be negative.
Since formation of 2-chloropropane has $\Delta H > 0$ while $-\Delta T\Delta S > 0$,
hence ΔG is always > 0 at all temperature, hence not spontaneous.

- (iv) 2-iodopropane can be converted to $(\text{CH}_3)_2\text{CHCO}_2\text{H}$ via a 2-step synthesis as shown.



Suggest the structure of compound X and give the reagents and conditions for each step. [3]

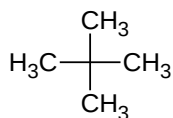
Compound X: $\text{CH}_3\text{CHCNCH}_3$

Step 1: ethanolic NaCN, heat

Step 2: $\text{H}_2\text{SO}_4(\text{aq})$, heat

- (d) Compound O, C_5H_{12} , is an alkane that reacts with chlorine gas to give only one monochlorinated product. [1]

- (i) Give the structural formula of compound O.



Compound O

Compound **P** is an isomer of **O**. Compound **P** reacts with chlorine to form several monochlorinated products but only one product **Q** contains a chiral centre. Under suitable conditions, **Q** forms two isomeric products **R** and **S**, C_5H_{10} .

(ii) Give the skeletal formulae of **P**, **Q**, **R** and **S**.

P	Q	R	S

- 2 (a) Strontium chloride, $SrCl_2$ is a typical salt which forms neutral aqueous solutions. It emits a bright red colour in a flame and this allows it to be used as a source of redness in fireworks.

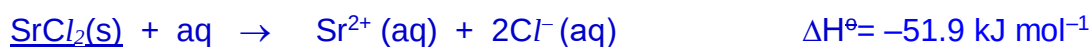
Use the data below to answer the questions that follow.

	/kJ mol ⁻¹
Standard enthalpy change of solution of $SrCl_2(s)$	-51.9
Standard enthalpy change of hydration of $Sr^{2+}(g)$	-1446
Standard enthalpy change of hydration of $Cl^{-}(g)$	-378

- (i) Define the term *standard enthalpy change of solution* of strontium chloride. [1]

Heat evolved when one mole of solid strontium chloride is dissolved in a solvent to form an infinitely dilute solution at 298 K and 1 bar.

Or



- (ii) Write an equation to show the relationship between the lattice energy of strontium chloride and ΔH°_{sol} of strontium chloride.

Hence, calculate the lattice energy of strontium chloride. [2]

$$\begin{aligned} \Delta H^{\circ}_{sol}(SrCl_2) &= -LE + \Delta H^{\circ}_{hyd}(Sr^{2+}_{(g)}) + 2\Delta H^{\circ}_{hyd}(Cl^{-}_{(g)}) \\ -51.9 \text{ kJ mol}^{-1} &= -LE + (-1446) + 2(-378) \\ LE &= -2150 \text{ kJ mol}^{-1} \end{aligned}$$

- (iii) Explain how the magnitude of enthalpy change of hydration of bromide ions differs from that of chloride ions. [2]

Compared to Cl^- , Br^- has larger radius and **lower charge/size**, resulting in a **weaker ion-dipole interactions** formed between Br^- and H_2O molecule.
 $\therefore \text{Br}^-$ has a smaller magnitude of ΔH_{hyd} .

- 2 (b) Mass spectrometry is an analytical technique which measures the masses of chemical species present in a sample. A mass spectrum can be used to determine the isotopic composition of a sample and to elucidate the chemical structure of a chemical compound.

The mass-to-charge ratio, also known as m/e value, is the M_r of chemical species present in the sample. The peak height ratio gives an indication of the isotopic ratio of an element in a sample.

In the first experiment, propene was reacted with liquid bromine in inert organic solvent. The organic products were then extracted and analysed using mass spectrometry. The mass spectrum is as shown in Figure 2.1.

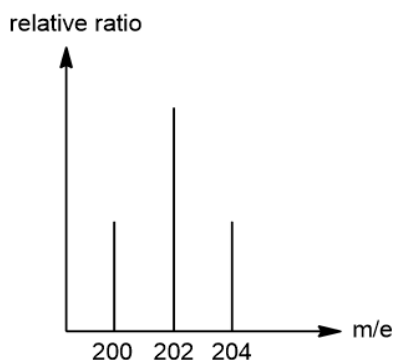


Figure 2.1. Mass spectrum of the products from reaction of propene with bromine

- (i) Draw the structure of the product formed when propene reacts with bromine in inert organic solvent. [1]

$\text{CH}_2\text{BrCHBrCH}_3$ (must draw)

- (ii) Based on the m/e values in Fig 2.1, deduce the two isotopes of bromine. [1]

^{79}Br and ^{81}Br

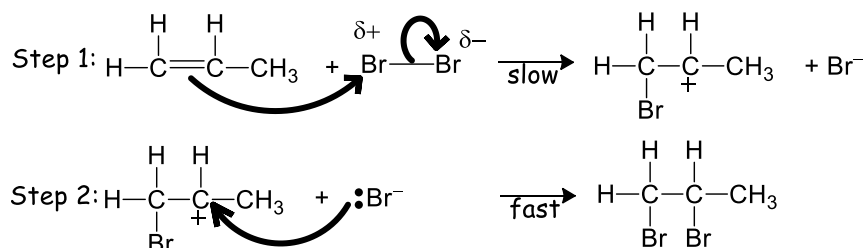
- (iii) Using your answer in (i) and (ii), determine the isotopic composition of bromine.

You may assume that carbon, hydrogen and oxygen only exist in one isotopic form of ^{12}C , ^1H and ^{16}O respectively. [1]

^{79}Br and ^{81}Br in the ratio of 1 : 1

- (iv) Describe the mechanism for the reaction between propene and bromine in inert organic solvent. Show relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs. [3]

Electrophilic addition



When propene is reacted with aqueous bromine, a different mass spectrum was obtained as shown in Fig 2.2. There are two additional peaks with greater peak height at m/e ratio of 138 and 140.

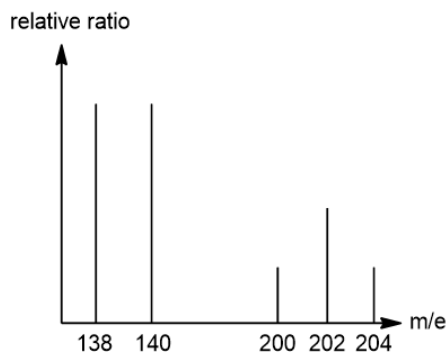


Fig 2.2 products from reaction of propene with aqueous bromine

- (v) Draw a possible structure that accounts for the product with m/e value of 138. [1]



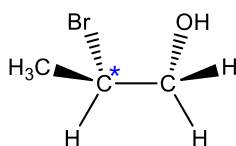
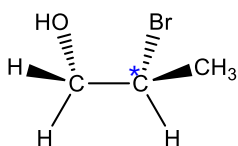
- (vi) Explain, with reference to the mechanism you have described in (iii), the m/e and relative ratio of the two additional peaks. [2]

Since the O atom of H_2O also possesses a lone pair of electrons, the O atom of H_2O can act as a nucleophile to attack the electron-deficient C^+ of carbocation, forming the halohydrin.

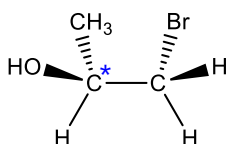
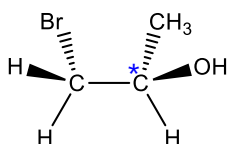
- water in large excess compared to Br_2 ,

- relatively much more H_2O molecule as nucleophile than the Br^- present in the mixture, hence the peak height at m/e = 138 is higher.

- (vii) Draw the four different possible products that are responsible for the peak at m/e value of 138. Copy the diagram below to your answer booklet and complete the diagram. [2]



Enantiomers of minor product



Enantiomers of major product

(viii) Ethanoic acid could be synthesised from propan-1-ol in two steps.

Suggest the structure of compound **Y** and give the reagents and conditions for each step.



Step 1: **excess conc** H_2SO_4 , heat

Or Al_2O_3 , heat

Step 2: KMnO_4 , dil H_2SO_4 , heat

Y: $\text{CH}_3\text{CH}=\text{CH}_2$

[3]

[Total: 19]

End of Paper